Isomorphic Questions

Multiple choice questions generated below. Answers in RED and incorrect answers in BLUE.

Q1. Which of the following statements is correct?

A. We can treat a cyclist as a perfect sphere when analyzing the detailed motion of the cyclist's ankle.
This is a poor approximation because ankle motion depends on the cyclist's shape and joint geometry, which a sphere cannot capture.

B. We can safely model a delivery truck as a point mass when studying how far it travels over several kilometers.
Over kilometer-scale position changes, the truck's size is negligible compared with the distance traveled, so a point-mass model is appropriate.

C. We must never assume motion occurs in a vacuum, even when we intentionally neglect drag forces.
Neglecting air resistance is often modeled by assuming vacuum-like conditions, so this "never" claim is too strong.

D. None of the above is correct.
Option B is correct, so "none of the above" cannot be correct.

Q2. Which of the following equations is correct?

A. 1 cm = 1 m.
This ignores the scale factor between centimeters and meters (1 m = 100 cm).

B. 1 = 0.01.
This confuses a conversion factor with an equality between dimensionless numbers.

C. 1 cm = 0.01 m.
Since 100 cm = 1 m, dividing by 100 gives 1 cm = 0.01 m.

D. 100 cm = 0.01 m.
This inverts the conversion because 100 cm equals 1 m, not 0.01 m.

Q3. Which option has the incorrect unit?

A. Pressure: pascal
Pascal (N/m^2) is the correct SI unit for pressure.

B. Momentum: kg · m/s
kg · m/s is the correct unit for linear momentum.

C. Power: joule
Power is energy per time, so its SI unit is watt (J/s), not joule.

D. Electric charge: coulomb
Coulomb is the correct SI unit for electric charge.

Q4. A scout drone is launched from a research outpost at the origin and detects a signal beacon at the coordinates $\vec{r} = -2\,\hat{\imath} + 7\,\hat{\jmath}$, where length units are in micromiles. Which of the following statements is correct?

A. The distance between the beacon and the outpost is 8 micromiles.
This comes from rounding incorrectly or assuming the components add linearly instead of using the Pythagorean theorem.

B. The distance between the beacon and the outpost is 7 micromiles.
This mistake treats the y-component magnitude as the full distance and ignores the x-component.

C. The distance between the beacon and the outpost is 2 micromiles.
This mistake treats the x-component magnitude as the full distance and ignores the y-component.

D. None of the above is correct.
The distance is $\sqrt{(-2)^2 + 7^2} = \sqrt{53} \approx 7.28$ micromiles, which is not 8, 7, or 2.

Q5. If the displacement of an object is given by the function $\vec{s} = \big((0.50\,\text{m/s}^3)t^3 - (1.25\,\text{m/s}^2)t^2\big)\hat{\imath}$, which of the following statement is correct?

A. This is a constant acceleration motion.
The acceleration depends on t after differentiating twice, so it is not constant.

B. The magnitude of the acceleration is 1.25 m/s^2.
This confuses the coefficient in the displacement term with the acceleration, which requires two time-derivatives.

C. The magnitude of the acceleration at $t = 0$ is 0 m/s^2.
Differentiating twice gives a nonzero constant term, so $a(0) \neq 0$.

D. The magnitude of the acceleration at $t = 0$ is 2.5 m/s^2.
Since $a(t) = d^2s/dt^2 = (3.0t - 2.5)\hat{\imath}$, we have $|a(0)| = 2.5$ m/s^2.

Q6. An object takes 3 second to move from position A ($\vec{r}_A = 3.0\,\hat{\imath} - 1.0\,\hat{\jmath}$) to position B ($\vec{r}_B = -2.0\,\hat{\imath} + 11.0\,\hat{\jmath}$) along a straight path, where length units are in meters. Which of the following statements is correct?

A. The moving distance is 13 m.
The straight-line distance is $|\Delta\vec{r}| = \sqrt{(-5)^2 + (12)^2} = \sqrt{169} = 13$ m.

B. The speed at $t = 0$ must be 4.33 m/s.
Only the average speed (distance divided by time) can be found from this information, not the instantaneous speed at $t = 0$.

C. The acceleration at $t = 0$ must be 1.44 m/s^2.
No information about how velocity changes with time is given, so acceleration cannot be determined.

D. None of the above is correct.
Option A is correct because the displacement magnitude is uniquely determined by the two position vectors.

Q7. A soccer ball is kicked with a speed v_0 at an angle θ above the horizontal. What is the speed of the ball when it is at the maximum height? (Neglect air resistance.)

A. v_0.
The speed is not generally the launch speed because the vertical component decreases to zero at the top.

B. $v_0 \cos\theta$.
At maximum height the vertical velocity is zero while the horizontal component remains $v_0 \cos\theta$, so the speed equals $v_0 \cos\theta$.

C. $v_0 \sin\theta$.
This is the initial vertical component, but at maximum height the vertical component is zero.

D. 0.
Only the vertical component becomes zero at the top; the horizontal component generally remains nonzero.

Q8. A drone is traveling with an initial velocity of $\vec{v}_0 = (15.0\ \text{m/s})\hat{\imath}$ when its thrusters engage, producing an acceleration of $\vec{a} = (-1.20\ \text{m/s}^2)\hat{\imath} + (0.80\ \text{m/s}^2)\hat{\jmath}$. Based on this scenario, which of the following statements is correct?

A. The x-component of the drone's velocity remains constant.
Because $a_x \neq 0$, the x-component of velocity must change with time.

B. The y-component of the drone's velocity remains constant.
Because $a_y \neq 0$, the y-component of velocity must change with time.

C. Both the direction of motion and speed change.
Nonzero acceleration components in both x and y change the velocity vector's components, altering both direction and speed.

D. None of the above is correct.
Option C correctly describes the effect of having acceleration in both coordinate directions.

Q9. If the magnitudes of both charges are doubled and their separation is increased by a factor of 4, what happens to the magnitude of the electrostatic force that each charge exerts on the other?

A. It stays the same.
This ignores that Coulomb's law scales as $q_1 q_2 / r^2$, so changing both q and r generally changes the force.

B. It decreases by a factor of 4.
Since $F \propto q_1 q_2 / r^2$, doubling both charges gives a factor of 4 while increasing distance by 4 gives a factor of 16 in the denominator, so F becomes $4/16 = 1/4$ of the original.

C. It increases by a factor of 4.
This accounts for doubling charges but forgets that increasing separation reduces force by the square of the distance factor.

D. It increases by a factor of 16.
This incorrectly treats the distance change as increasing the force rather than decreasing it via r^2.

E. It decreases by a factor of 2.
This is a common arithmetic slip that does not correctly combine the $\times 4$ charge effect with the $\div 16$ distance-squared effect.

Q10. A probe can escape a celestial body of radius R and mass M at an escape speed given by $v = \sqrt{\frac{2GM}{R}}$, where G is the universal gravitational constant. Based on this expression for escape speed, which of the following statements is correct?

A. A more massive probe requires a higher speed to escape from the same body.
The probe's mass does not appear in $v = \sqrt{2GM/R}$, so escape speed is independent of the escaping object's mass.

B. A less massive probe requires a lower speed to escape from the same body.
This repeats the misconception that escape speed depends on the object's mass, which it does not in this formula.

C. Escape speed is higher for a body with smaller mass and larger radius.
Because $v \propto \sqrt{M/R}$, decreasing M and increasing R both make v smaller, not larger.

D. Escape speed is higher for a body with greater mass and smaller radius.
Since $v \propto \sqrt{M/R}$, increasing M and decreasing R both increase the escape speed.

Q11. The force $\vec{F} = (-2.50\,\text{N})\hat{\imath} + (6.00\,\text{N})\hat{\jmath}$ acts on a particle as it moves along the y axis from $y = 1.20$ m to $y = 4.70$ m. What is the work, in joules, done by the force on the particle?

A. 21.0 J.
The displacement is $\Delta\vec{r} = (3.50\text{ m})\hat{\jmath}$, so $W = \vec{F}\cdot\Delta\vec{r} = (6.00)(3.50) = 21.0$ J.

B. −8.75 J.
This incorrectly uses the x-component of force even though the displacement is purely along y.

C. 8.75 J.
This takes the magnitude of the x-component contribution, but there is no x-displacement so the x-force does no work.

D. 27.5 J.
This commonly comes from using the force magnitude times distance instead of the dot product with the correct direction.

Q12. A 3.0 kg cart initially moving at 4.0 m/s collides with a bumper and rebounds at a speed of 1.5 m/s moving backward. The collision lasts for 0.25 s. What is the average force exerted on the cart during the collision?

A. 66 N.
Taking the initial direction as positive, $\Delta p = m(v_f - v_i) = 3.0(-1.5 - 4.0) = -16.5\ \text{N}\cdot\text{s}$, so $|F_{\text{avg}}| = |\Delta p|/\Delta t = 16.5/0.25 = 66$ N.

B. 16.5 N.
This forgets to divide the impulse by the collision time to get average force.

C. 132 N.
This typically comes from doubling the correct force by mishandling the sign change or using the wrong time interval.

D. None of the above is correct.
Option A is correct because it correctly applies impulse-momentum over the given time.

Q13. An 85 kg catcher is initially at rest. A 145 g baseball traveling at 90 mph is caught and held in the glove (the ball and catcher move together immediately after the catch). What is the speed of the catcher after catching the ball?

A. 0.069 m/s.
Using momentum conservation, $v = \frac{m_b v_b}{m_c + m_b} = \frac{0.145\times(90\text{ mph}\approx 40.2\text{ m/s})}{85+0.145} \approx 0.069$ m/s.

B. 0.83 m/s.
This typically results from forgetting to divide by the catcher's mass (treating $m_b v_b$ as a speed).

C. 40 m/s.
This assumes the catcher keeps the ball's initial speed, ignoring the much larger catcher mass in an inelastic capture.

D. 28 m/s.
This often comes from converting mph incorrectly or applying an unrelated kinetic-energy idea instead of momentum conservation.
